Cover Page

Project Name: Event Scheduling & Planning System

TA Name: Mohamed Haroun

Team ID: 83

Member Informations

Member Name	Seat Number	Department
اسلام هانى حسين عيد	2023170090	SC
معاذ ثروت البكرى مطاوع	2023170621	
عبد الرحمن بهاء مصطفى	2023170330	
احمد اسماعيل حامد السيد	2023170019	
يوسف جورج أمين سليمان	2023170722	
عمرو محمد عمرو محمود	2023170408	
عمر احمد عبده صالح	2023170381	

SRS Document

(83) - Event Scheduling and Planning System

Mohamed Haroun

SC -معاذ ثروت البكرى مطاوع , SC -اسلام هانى حسين عيد

SC -احمد اسماعيل حامد السيد , SC -عبد الرحمن بهاء مصطفى

SC -عمرو محمد عمرو محمود , SC -عمر احمد عبده صالح

SC -يوسف جورج أمين سليمان

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1. Introduction

The Event Scheduling & Planning System is a digital platform designed to automate event management and optimize resource allocation. It addresses the challenges of manual scheduling by providing a structured workflow for event creation, administrative approval, and user registration.

By offering role-specific features for Attendees, Organizers, and Admins, the system ensures that time slots are managed efficiently without overlaps. Key functions include event discovery, automated notifications, and attendee tracking. This integrated approach reduces administrative overhead and enhances communication, creating a more organized and connected environment for all users.

2. User Requirements

- The attendees and organizers should be able to create an account and log in.
- The attendees should be able to view all approved events.
- The attendees should be able to search (by name) and filter events (date).
- The attendees should be able to view event details (date, time, description).
- The attendees should be able to register for events.
- The attendees should be able to cancel event registration.
- The organizers should be able to create events.

- The organizers should be able to update event details.
- The organizers should be able to cancel their "Pending" events; however, canceling an "Approved" event requires administrative notification/audit.
- The organizers should be able to view list of attendees.
- The organizers should be able to view all available time slots to select for event creation.
- The organizers should be able to ask admin for approving notifing events.
- The admin should be able to approve organizers events for a given time slots.
- The admin should be able to create time slots for events.
- The admin should be able to view all events with its time slots.
- The admin should be able to broadcast a notification to all attendees regarding the details of a newly approved event.

3. Functional Requirements

Functional Requirement 1: User Account Management

- Description/Action: The system shall allow users to create a new account and authenticate (log in) to access role-specific features.
  - For Registration: Name, Username, Password, Age, Gender, Role (Attendee or Organizer)
  - For Login: Username, Password
- Source: Attendee or Organizer.
- Pre-condition: User must not have an existing account (for registration); user must be registered (for login).
- Post-condition:
  - For registration: A new user record is created in the database with a unique User ID automatically generated by the system, and the user's details are saved
  - For login: A secure session is established using the existing user record and the user is redirected to their dashboard.
- Output: Success/Error message and redirection to the dashboard.

Functional Requirement 2: Event Search and Filtering

- Description/Action: Enables attendees to browse events using text searches and specific criteria.

- **Requirements/Inputs: Event name , Date.**
- **Source: Attendee.**
- **Pre-condition: User is authenticated.**
- **Post-condition: The system queries the database for matching "Approved" events.**
- **Output: A filtered list of events matching the search parameters.**

Functional Requirement 3: Registration & Cancellation

- **Description/Action: Allows attendees to join an event or withdraw their participation.**
- **Requirements/Inputs: Event ID, User ID.**
- **Source: Attendee.**
- **Pre-condition: For registration: Event must have available capacity. For cancellation: User must be currently registered.**
- **Post-condition: The registration table is updated, and the event capacity count is adjusted.**
- **Output: Confirmation message of registration or withdrawal.**

Functional Requirement 4: Event Creation & Submission

- **Description/Action: Organizers define event parameters and submit them for administrative review.**
- **Requirements/Inputs: Title, Description, Date, Time Slot, Capacity.**
- **Source: Organizer.**
- **Pre-condition: Organizer is logged in; the selected time slot must be "Available."**
- **Post-condition: A new event record is created with a "Pending" status.**
- **Output: Confirmation that the event has been sent to the Admin for approval.**

Functional Requirement 5: Event Detail Maintenance & Cancellation

- **Description/Action: Organizers can modify details of their events or withdraw a submission.**
- **Requirements/Inputs: Event ID, updated field values.**
- **Source: Organizer.**

- **Pre-condition: The organizer must be the owner of the specific Event ID. If the event is already "Approved," the system must trigger a notification to the Admin upon cancellation.**
- **Post-condition: Event records are updated, or the status is changed to "Canceled," and the associated time slot is released back to "Available."**
- **Output: Success message and updated organizer dashboard.**

Functional Requirement 6: Resource Oversight (Role-Based Visibility)

- **Description/Action: The system displays scheduling data filtered by the user's role to ensure clarity and prevent clutter.**
- **Requirements/Inputs: User Role (Organizer/Attendee).**
- **Source: System Database.**
- **Pre-condition: User is authenticated.**
- **Post-condition: Read-only action.**
- **Output:**
    - **Organizers: A calendar view of all "Available" time slots for planning.**
    - **Attendees: A list/view of "Approved Events" only (Attendees do not see raw "Slots").**

Functional Requirement 7: Time Slot Management

- **Description/Action: Admin defines the operational hours and creates the "blocks" that organizers can book.**
- **Requirements/Inputs: Date range, Start/End time, Slot duration.**
- **Source: Admin.**
- **Pre-condition: Admin is authenticated.**
- **Post-condition: The system's master schedule is populated with "Available" slots or "Blackout" periods.**
- **Output: A visual calendar showing the updated master schedule.**

Functional Requirement 8: Event Approval & Booking

- **Description/Action: Admin reviews "Pending" events to either authorize them or reject them.**

- Requirements/Inputs: Event ID, Decision (Approve/Deny).
- Source: Admin.
- Pre-condition: An event is in "Pending" status.
- Post-condition:
  - If approved: the system automatically generates a unique Event ID, the event is permanently saved in the database with that ID, and the time slot becomes Booked.
  - If denied: event data is discarded and time slot remains Available.
- Output:
  - if approved: organizer receives approval notification and event becomes visible to attendees.
  - If denied: organizer receives rejection notification and no record is saved."

Functional Requirement 9: Multi-Tier Notifications

- Description/Action: The system handles two levels of communication: automated status updates for organizers and event broadcasts for attendees.
- Requirements/Inputs: Event ID, Approval Status.
- Source: Automated (Triggered by Admin Action).
- Pre-condition: Admin has finalized the approval/rejection of an event.
- Post-condition:
  - 1. System Notification: Automatically sent to the Organizer confirming the status change.
  - 2. Broadcast (Automated): A "New Event" alert is sent to all Attendees once an event is "Approved."
- Output: Notification alerts appearing in user dashboards and "New Event" tag added to the public event list.

4. Non-Functional Requirements

- Performance

  The system shall load all pages within 3 seconds under normal operating conditions. Search queries submitted by the user, including searching by event name shall return results within 2 seconds. This ensures that attendees, organizers, and admins can interact with the system efficiently without unnecessary delays.

- **Security**

  All user passwords shall be encrypted using a standard hashing algorithm before being stored in the database. The system shall enforce role-based access control, ensuring that attendees, organizers, and admins can only access the pages and features permitted for their role.

- **Development Constraints**

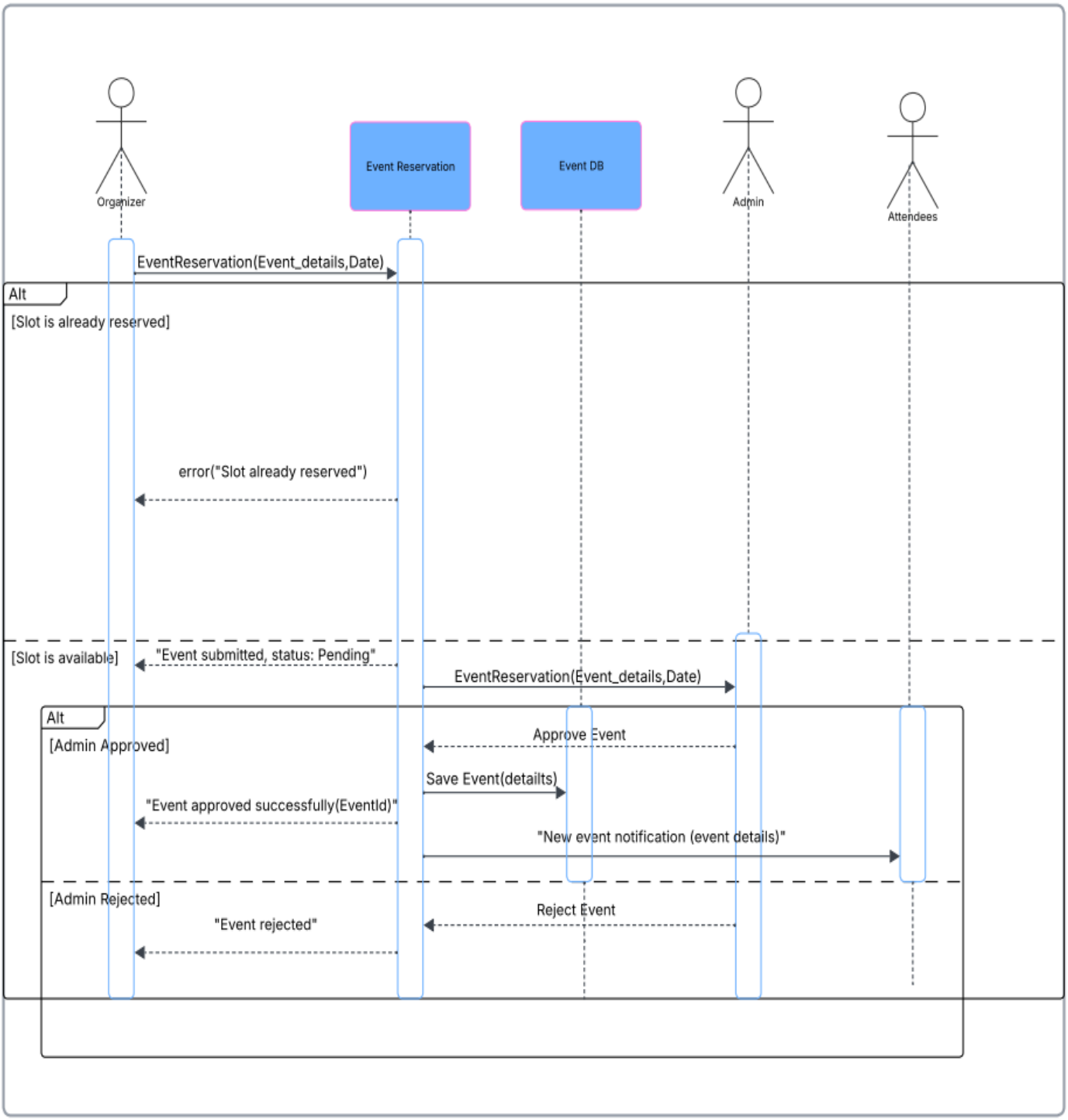
  The system shall be developed using C# with the .NET framework inside Visual Studio as the designated IDE for the entire development team. The development process shall follow an incremental development approach, allowing the system to be built, tested, and reviewed in stages throughout the project timeline. All source code shall be version-controlled and managed using GitHub, ensuring that all team members can collaborate, track changes, and maintain a consistent and up-to-date codebase throughout the project.

- **Usability**

  The system shall provide a simple and clean interface that allows any user whether an attendee, organizer, or admin to navigate and use the system without the need for any prior training or technical knowledge. A new attendee shall be able to create an account, search for an event, and complete their registration in no more than 5 minutes from their first visit to the system.

- **Maintainability**

  The system source code shall be written in a clean and organized manner, with clear comments added to 100% of all major functions and modules to ensure that any team member can understand and modify the code without difficulty. All changes and updates to the code shall be committed and documented on GitHub so that the full history of modifications is traceable at any point during the project.

Use Case Diagram

Sequence Diagram